# Mensik_Experiments_PS_7

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## Chapter 6 Pset

### Problem 1

The first quantity describes the expected treatment status for subjects assigned to treatment who take the treatment. The second quantity describes the treated potential outcome for compliers and with two sided NC, always takers. The third quantity describes the treated outcome (expected value of) for always takers.

### Problem 3a

Yes, the intent-to-treat effect conditional on compliers (d = z) is the ATE.

#### Problem 3b

False: Among defiers d = inverse(z), so the ITT is the inverse of the ATE.

#### Problem 3c

Neither? The ATE for always takers and never takers is an unknown quantity. So, it is false to say their ITT = ATE.

### Problem 5

30 pct always takers, 40 pct never takers, 15 pct compliers, 15 pct defiers, what is ITTD: 0 because 30 ATs and NTs don't matter and 15-15 = 0.

### Problem 6

Monotonicity is violated and sample has compliers and defiers. pi_c is proportion of subjects compliers pi_d is proportion defiers

show CACE is still identified if ATE among defiers is ATE among compliers and the proportions are not equal

Monotonicity means pi_d is not 0

$$ITT = ATE|Complier * \pi_C - ATE|Defier * \pi_D$$

ATE|Compliers = ATE|Defiers, so $ITT = (ATE|Complier \text{ (or) } Defier) * (\pi_C - \pi_D)$ .

Defiers means the ITT_D is the proportion of compliers net the proportion of defiers, so:

$$CACE = \frac{ITT_Y}{ITT_D} = \frac{(ATE|CorD) * (\pi_C - \pi_D)}{\pi_C - \pi_D} CACE = ATE|CorD$$

## Problem 8

(Pretending this extremely problematic experiment is not real)

### a

Noninterference assumes that the outcomes of one household (calls to a hotline and later arrest) are only dependent on whether the police issued a warning or arrested the accused.

This does not seem plausible in a context (anywhere in the US, Milwaukee very much included) where crime, poverty, and policing/coercion are raced and classed.

Moreover, I cannot imagine that police would arrest (whether holding overnight or limited time) or only issue a warrant only based on instructions from a researcher. Rather, it seems wholly probable that a frustrated or badly (un)trained cop would escalate and arrest someone in part because they had left someone else off with a ticket. So no – I do not see how noninterference is possible.

Excludability assumes that outcomes are affected by the treatment. A cop who is "assigned to arrest" may very well behave far more aggressively, detain someone for longer regardless of circumstances (for example – the accused misses hourly wages, or is in precarious employment where absence/lateness has consequences, or on probation/parole so arrest means re-sentencing/re-incarceration). The well-studied collateral consequences of this terrible experiment mean there is no reason to think that future outcomes are conditional on treatment alone.

There will absolutely be police who do the opposite of what they were instructed to do. Not all cops, (!) but undoubtedly many are inured to being in positions of power, and getting told what to do by some ivory tower academics who do not understand it seems much of anything about their subject seems a sure trigger for little acts of rebellion.

### b

Always takers show up in the control group, and randomization lets us assume that pi_alwaystakers(d=0) = pi_alwaystakers(d=1)

```
ITT_D <- (789/802)-(2/398)

ITT_calls <- (597/802)-(261/398)

CACE_calls <- ITT_calls/ITT_D

itt_arrests <- 303/802 - 151/398

cace_arrests <- itt_arrests/ITT_D

ITT_D
```

```
## [1] 0.9787654
```

```
ITT_calls
```

```
## [1] 0.08861013
```

```
CACE_calls
```

```
## [1] 0.09053256
```

```
itt_arrests
```

```
## [1] -0.001591499
```

```
cace_arrests
```

```
## [1] -0.001626027
```

The compliance rate is nearly 1, so the ITT for calls and CACE are similar – assignment to treatment and treatment receipt are associated with an increase in the outcome (911 calls) of 8.9 and 9.0 respectively.

An initial arrest rather than warning is associated with a lower likeyhood of later arrest for compliers.

**c**

Not assuming monotonicity means we throw back in the possibility of defiers (d(z=1) = 0, d(z=0)=1)) so the ITTD is not just the proportion of compliers - 0 (by assumption). Each observed treatment status group has mix of possible compliance types.

We want to know the proportions:

P(d = 0 | z = 1) = 3+1/404+398 = 4/802 P(d = 1 | z = 1) = 404+398 = 798/802 P(d = 0 | z = 0) = 396/398 P(d = 1 | z = 0) = 1+1 / 398 = 2/398

P(d = 0 | z = 1) = P(NT) + P(D) = 408

The upper bound for never-takers occurs when defiers = 0. The max possible P(NT) then = 4/802

Min bound for never-takers = 0.

P(d = 1 | z = 0) = P(AT) = P(D) = 2/398

This is the max possible proportion of always-takers.

Min possible proportion of always-takers means max defiers, which we have as 4/802. So min always takers is (2/398-4/802).

```
2/398-4/802
```

```
## [1] 3.759446e-05
```

P(d = 0 | z = 0) = P(Complier)+P(NeverTaker) = 396/398.

Compliers are minimum value when never takers are maxed (NT = 4/802)

So min compliers is: 396/398 - 4/802 Max compliers: 396/398.

Substantively, it looks like around 99% of subjects are compliers, and 0.905% are defiers.

Removing the monot. assumption leads to an increase in assumed proportion of defiers and compliers.

**d**

Possible types when warning assigned (`d_0 = 1`)

| d_1 | d_2 |
|---|---|
| 0 | 0 |
| 2 | 2 |
| 1 | 1 |
| 0 | 1 |
| 0 | 2 |
| 1 | 0 |
| 2 | 0 |
| 1 | 2 |
| 2 | 1 |

Possible types when short arrest assigned (`d_1 = 1`)

| d_0 | d_2 |
|---|---|
| 0 | 0 |
| 2 | 2 |
| 1 | 1 |
| 0 | 1 |
| 0 | 2 |
| 1 | 0 |
| 2 | 0 |
| 1 | 2 |
| 2 | 1 |

Possible types when `d_2 = 1`

| d_0 | d_1 |
|---|---|
| 0 | 0 |
| 2 | 2 |
| 1 | 1 |
| 0 | 1 |
| 0 | 2 |
| 1 | 0 |
| 2 | 0 |
| 1 | 2 |
| 2 | 1 |

**e**

There are 10 types.

## Problem 10

### Problem 10a

Monotonicity is plausible. The exception might be if its known that a polling station will have observers, people who might get up to shenanigans would move their tricks to another polling location, and observers would get wind of it and independently decide to go to those polling stations instead.

That seems like a stretch, but perhaps more likely from the perspective of a poll watcher assigned to one location who happens to learn that another has more issues – but that would not seem systematic or a consequence of assignment.

### Problem 10b

Proportion never-takers 341/409 = 83.37%

Proportion always takers: 21/1562 ~ 1.34%

Proportion compliers: 68/409 ~ 16.62% - 1.3 = 15.32%

### Problem 10c

Non-interference means that assigning election observers (d = 1 and z = 1) to a polling location does not affect the outcome (number of ballots declared) at another location. That wouldn't hold if voter fraud etc just concentrates elsewhere.

### Problem 10d

```
getwd()
```

```
## [1] "/Users/kom11/Documents/GitHub/regression-study/Experiments"
```

```
data_10 <-read.csv("/Users/kom11/Documents/GitHub/regression-study/Experiments/Hyde_POP_2012.csv")

ITT <- mean(data_10$invalidballots [data_10$Sample == 1]) - mean(data_10$invalidballots [data_10$Sample

ITT_D <- (sum(data_10$observed [ data_10$Sample == 1])) / (sum(data_10$Sample == 1)) - (sum(data_10$obs

CACE <- (ITT / ITT_D)

ITT
```

```
## [1] 4.824097
```

```
CACE
```

```
## [1] 31.56824
```

The ITT is 4.28 and CACE 31.65.

**Problem 10e**

```
set.seed(6081709)

declaration <- declare_ra(
  N = nrow(data_10),
  m = sum(data_10$Sample)
)

declaration
```

```
## Random assignment procedure: Complete random assignment
## Number of units: 1971
## Number of treatment arms: 2
## The possible treatment categories are 0 and 1.
## The number of possible random assignments is approximately infinite.
## The probabilities of assignment are constant across units:
##    prob_0    prob_1
## 0.7924911 0.2075089
```

```
permmatrix <- obtain_permutation_matrix(declaration)

estimates_vector <- rep(NA, 10000)

for(i in 1:10000){
  data_10$ri <- permmatrix[, i]

  estimates_vector [i] <- difference_in_means(invalidballots ~ ri, data = data_10)$coefficients[1]

}

mean(abs(estimates_vector) >= abs(ITT))
```

```
## [1] 0.487
```

Testing the null that the ITT is 0 for all subjects serves the same purpose as testing that the ATE's 0 for all compliers because a 0 ITT means a 0 CACE.

The high p value (0.487) means we fail to reject the null that there is no ITT effect for any poll. We can say that assignment of election observers did not have a significant effect on the number of valid ballots officialls report.